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Title: A novel *RELA* truncating mutation in familial Behçet's Disease-like mucocutaneous ulcerative condition.

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ABSTRACT

Objectives: Monogenic Behçet's Disease (BD)-like conditions are increasingly recognised and to date predominantly involve loss-of-function variants in *TNFAIP3.* The objective of this study was to identify genetic and pathobiological mechanisms associated with a BD-like mucocutaneous ulcerative syndrome and neuromyelitis optica (NMO) occurring in a three-generation Irish family (n=5 cases; n=5 familial controls).

Methods: Whole exome sequencing was used to identify potential pathogenic variants in affected family members and determine segregation between affected and unaffected individuals. Relative RELA expression in peripheral blood mononuclear cells was compared by western blot. Human epithelial and *RelA*$^{-/-}$ mouse fibroblast experimental systems were used to determine the molecular impact of the RELA truncation in response to TNF. NF-κB signalling, transcriptional activation, apoptosis and cytokine production was compared between wild-type and truncated RELA in experimental systems and patient samples.

Results: A heterozygous cytosine deletion at position c.1459 in RELA was detected in affected members. This mutation results in a frameshift p.His487ThrfsTer7, producing a truncated protein disrupting two transactivation domains. The truncated RELA protein lacks a full transactivation domain. The RELA protein variants were expressed at equal levels in peripheral mononuclear cells. *RelA*$^{-/-}$ mouse embryonic fibroblast (MEF) cells expressing recombinant human RELAp.His487ThrfsTer7 were compared to those expressing RELA^{WT}, however there was no difference in RELA nuclear translocation. In *RelA*$^{-/-}$ MEF expression of RELAp.His487ThrfsTer7 resulted in 1.98-fold higher ratio of cleaved caspase-3/caspase-3 induced by TNF (p=0.036) compared to RELA^{WT}.

Conclusions: Our data support *RELA* loss-of-function mutations as causing BD-like autoinflammation and NMO via impaired NF-κB signalling and increased apoptosis.

KEYWORDS

Behcet's Disease, Neuromyelitis Optica, autoinflammation, monogenetic, RELA, apoptosis.

INTRODUCTION

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Behçet's disease (BD) is an autoinflammatory vasculitic condition characterised by oral and genital ulceration, uveitis, rashes such as erythema nodosum and arthralgia. The aetiology is multifactorial with a significant genetic component and shared susceptibility loci with Crohn's disease and leprosy implicating mucosal factors and the innate immune response in BD risk(1).

Nuclear factor-κB (NF-κB) is a family of inducible transcription factors, which act as master regulators of immune and apoptotic signalling by controlling the expression of a large array of genes. The NF-κB family is composed of five structurally related members; NF-κB1 (p50), NF-κB2 (p52), RELA (p65), RELB and c-REL. These proteins mediate transcription of target genes by binding to a κB enhancer DNA element as hetero- or homo-dimers(2).

Autosomal dominant forms of BD have been linked to loss-of-function (LOF) mutations in TNFAIP3 (A20), a negative regulator of NF-κB (3). Differences in clinical features compared with BD include onset in early childhood, recurrent fevers, severe ocular disease, colitis and polyarthritis(4). A single family has been reported with an autosomal dominant early onset mucocutaneous ulcerative syndrome caused by a RELA mutation (5). These studies implicate distinct genetic and cellular mechanism involving the NF-kB pathway in familial forms of BD.

Like BD, neuromyelitis optica (NMO/Devic's Disease) is a relapsing and remitting inflammatory disorder. NMO is confined to the central nervous system and is associated with specific anti-aquaporin-4 (AQP4) antibodies. Several studies have demonstrated that both conditions share similar pathophysiology, and NF-κB is key(6). Proteasome and immunoproteasome inhibitors that prevent nuclear accumulation of NF-κB have demonstrated profound effects on astroglial responses to NMO IgG(7).

Despite being the hallmark feature of NMO, optic neuritis is also a recognized rare manifestation of BD. This feature, sometimes described as a posterior segment ocular involvement, is part of the International Study Group for BD (ISGBD) criteria and the revised International Criteria for BD (ICBD)(8).

In this study we determined the genetic and molecular basis of an autosomal dominant mucocutaneous ulcerative syndrome and NMO in five members of a three-generation Irish

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family. Whole exome sequencing identified a mutation in *RELA* resulting in expression of a truncated RELA protein, with suppression of RELA-dependent transactivation activity and increased apoptosis.

METHODS AND MATERIALS

Public and Patient Involvement

This study has a patient research partner member of the project advisory group who mentors the researchers in lay communication and advises on patient consent forms.

Ethics and sample collection

Written informed consent was obtained from all participants or guardians. This study was approved by University Hospital Limerick's local ethics committee in accordance with the Declaration of Helsinki. Whole blood or saliva were obtained from all family members. PBMCs were obtained from n=3 familial controls and n=4 heterozygotes.

Whole exome sequencing

Libraries were sequenced on an Illumina HiSeq 3000 (Illumina, USA). Variants were annotated using Alamut-HT high-throughput annotation software. The identified variant was validated by Sanger sequencing. For expanded methods see Supplementary Methods.

Cell culture

PBMCs were isolated from whole blood using density gradient centrifugation. HEK293T cells (ATCC) were cultured in Dulbecco's modified Eagle medium supplemented with 10% heat-inactivated fetal bovine serum. MEF cells lacking NF-κB subunit p65 (*RelA*$^{-/-}$) were a gift from Alexander Hoffman (UCLA, CA, USA). Recombinant human or mouse TNF (PeproTech, USA) was used at 10ng/ml.

RT-PCR

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Gene expression was measured via TaqMan real-time using delta CT relative quantitation. Assay IDs are in Supplementary Methods.

Plasmid Transfection:

pRelA-EGFP plasmid (a generous gift from Prof. Michael White, University of Manchester) was used as a template to generate RELAp.His487ThrfsTer7 and $RELA^{WT}$ plasmids and confirmed by capillary sequencing. Transfection of DNA plasmids were performed using Lipofectamine™ 3000 transfection reagent (Thermo Fisher Scientific, USA) according to manufacturer's instructions. For expanded methods see Supplementary Methods.

Luciferase Reporter Assay

NF-κB-Luciferase PEST reporter vector (Promega, USA) plus the Renilla luciferase control reporter vector (pRL-SV40; Promega, USA) were co-transfected into *RelA*$^{-/-}$ MEF cells. Cells were incubated with 10ng/ml TNFα for 5 hours and luciferase activity measured using the dual-glo luciferase assay system (Promega, USA). For expanded methods see Supplementary Methods.

Western blot

Cells were treated with 10ng/ml TNFα for 24 hours and whole cell lysates extracted. Electrophoresis was carried out using 12% Tris-Glycine SDS-PAGE and wet transfer to polyvinylidene difluoride membranes (Thermo Fisher scientific, USA). Membranes were blocked with 5% protein and antigen–antibody complexes were visualized using chemiluminescence. For expanded methods see Supplementary Methods.

Cytokine assays

ELISAs were performed to detect human IL6, TNF and mouse CXCL10 and IL6 protein concentrations (R&D Systems, USA) and performed as per manufacturer's instructions using 100 μL of supernatant per well.

Immunofluorescence

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RelA$^{-/-}$ MEF cells (n=17,500) were seeded into a Millicell EZ 8-well chamber slides (Millipore, USA). Cells were fixed in 4% Paraformaldehyde (Sigma-Aldrich, USA) for 20 min, washed 3 times with PBS, permeabilised with 0.1% Triton X-100 (Sigma-Aldrich, USA) 5 min, washed three times with PBS, and incubated in blocking solution containing 0.5% BSA (Sigma-Aldrich, USA), 50mM ammonium chloride in PBS for 30 min. Cells were probed with primary antibody (1:1,000) in blocking solution containing 0.02% sodium azide (Sigma-Aldrich, USA) for 1 hour, washed 3 times with PBS, and probed with Goat-α-rabbit Alexa Fluor-568 (1:400, Life Technologies) diluted in blocking solution for 30 min. Cells were washed and coverslip was mounted using Fluroshield™ with DAPI mounting media (Sigma-Aldrich, USA).

Confocal microscopy

Confocal images were acquired on a FluoView FV1000 laser scanning microscope (Olympus, Japan). Images were acquired with a 60x/1.35 NA UPlanSApo oil immersion objective (Olympus, Japan) at a resolution of 1024 × 1024 pixels in sequential scanning model.

Statistical analysis

Data are presented as mean ± SEM. Data was graphed to assess distribution and transformed where required to meet parametric assumptions. A Student's t test was used to compare two groups; using the unpaired test for independent groups, and type-1 test for paired samples. To compare more than two groups, ANOVA with Tukey's HSD post-hoc test was used. Statistical analysis was performed IBM SPSS Statistics Version 24. An alpha value of 0.05 was employed.

RESULTS

Clinical Phenotype

This study involved an Irish Caucasian family of ten individuals (Fig 1A). The index case (I.2) reported a transient period of recurrent severe oral ulcers as a teenager, his symptoms however spontaneously resolved over 30 years ago. Two half-sisters (II.2 and II.5) were diagnosed with early-onset Behçet's disease satisfying the ISGBD criteria at 10 and 15 years old respectively. The illness was characterized by recurrent oral aphthosis, genital aphthosis and skin pustulosis. In

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contrast to cases in the literature with A20 haploinsufficiency(4), affected individuals in this study were ANA negative. A third sister (II.3) developed neuromyelitis optica (NMO), characterized by recurrent optic neuritis with positive AQP4 antibody. Of note serial imaging demonstrated no evidence of transverse myelitis or cerebral involvement. A son of the index case declined to be involved in the study but was reported to be well and asymptomatic. Subsequently, a daughter of one of the BD sisters (III.1) developed recurrent oral aphthosis at the age of 10-years. The 2 half-sisters with BD were successfully treated with anti-TNF medication (Etanercept) and the patient with NMO with anti-CD20 (Rituximab). The demographic, clinical manifestations, and the HLA-B*51 status of all family members were recorded (Supplementary Table 1).

Identification of a novel *RELA* variant

Whole exome sequencing identified a heterozygous deletion c.1459delC in exon 11 of *RELA* (ENST00000406246), this was validated by Sanger sequencing and was detected in all affected individuals and the individual with NMO (Fig 1B). The mutation resulted in a frameshift encoding RELA amino acid 487 followed by 5 amino acids and a stop codon (QLSPC*), encoded a truncated RELA protein of 492 amino acids, approximately 6kDa smaller than wild-type RELA (Fig 1C). This mutation has not been identified in publicly available genetic datasets (https://gnomad.broadinstitute.org).

Individuals heterozygous for *RELA* c.1459delC express RELAp.His487ThrfsTer7 protein

Individuals heterozygous for *RELAc.1459delC* have two bands when probed with an anti-RELA antibody targeted to an N-terminus epitope, with an approximate 6kDa difference in protein size, as predicted (Fig 2A). Heterozygotes and familial homozygous wild-type (WT) controls were also probed with a C-terminus antibody targeted to transactivation domain 1 which was predicted to be missing from RELAp.His487ThrfsTer7. As expected, only a single band (65 kDa) is detected in all samples (Fig 2A).

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NF- κB pathway demonstrates altered expression and sensitivity to TNF in RELAp.His487ThrfsTer7 individuals

Compared to WT familial PBMC controls, RELAp.His487ThrfsTer7 heterozygotes had higher basal RELA (32.6-fold upregulation, p=0.011 Fig 2B), NFKBIA (10.5-fold upregulation, p=0.002 Fig2B) and TNFAIP3/A20 (24.8-fold upregulation, p=0.011 Fig 2B) gene expression.

However RELAp.His487ThrfsTer7 patient PBMCs exhibited reduced TNF-induced induction compared with WT PBMCs, with RELA downregulated 2.8-fold (p=0.037); NFKBIA downregulated 2.7-fold (p=0.012; and TNFAIP3 downregulated 2.705-fold (p=0.035) compared to basal (unstimulated) cells (Fig 2B).

Transcriptional activity is impaired in RELAp.His487ThrfsTer7

To determine if the transcriptional activity of the two RELA isoforms differed in response to TNF we performed a NF-κB luciferase experiment. Transfection efficiency was comparable between $RELA^{WT}$ and RELAp.His487ThrfsTer7 transfections (Fig S3). MEF $RelA^{-/-}$ expressing $RELA^{WT}$ had 6.78-fold (+/-0.78, p=0.003) increased promoter activity compared to those expressing RELAp.His487ThrfsTer7(Fig 2C).

Nfkbia gene expression was measured in these cells (Fig S2). RELAp.His487ThrfsTer7 had reduced expression compared to $RELA^{WT}$ basally ($p<0.001$) and in response to TNF (p=0.009).

RELAp.His487ThrfsTer7 does not impede RELA translocation to the nucleus

To determine if the differential NF-κB-responsive promoter activity was related to translocation to the nucleus of the RELA variants we used the *RelA*$^{-/-}$ MEF transfection model as before. There

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was no significant difference in translocation (Fig 2D), with both demonstrating nuclear translocation at 10 minutes post-TNF exposure.

RELAp.His487ThrfsTer7 and TNF-induced cytokine production

Of the three symptomatic RELAp.His487ThrfsTer7 individuals, only one had detectable basal mRNA levels of IL6, whereas samples from all three WT familial controls had detectable IL6 (Fig S1A). Gene expression of the chemokine CCL5 was upregulated 15.2-fold in RELAp.His487ThrfsTer7 patient PBMCs compared to WT familial controls, however TNF induced downregulation of CCL5 was not statistically significant (Fig S1B). There was no significant difference in TNF secretion between patient and familial control PBMCs (p=0.778, Fig S1C). In *RelA*$^{-/-}$ MEF, we observed no difference in the secretion of IL6 or CXCL10 between RELA^{WT} and RELAp.His487ThrfsTer7 transfected cells (Fig S1D and E).

RELAp.His487ThrfsTer7 increases apoptosis

Exposure to TNF increased expression of the pro-apoptotic protein BAX under all conditions in HEK293T cells (P<0.001, Fig 3A). Compared to those transfected with RELAp.His487ThrfsTer7, HEK293T cells transfected with RELA^{WT} had reduced expression of BAX (basal p<0.001; TNF p<0.001) and increased expression of the anti-apoptotic protein BCLx (BCL2L1) (basal p=0.011; TNF p<0.001) , there was no difference in BCL2 expression between RELA^{WT} and RELAp.His487ThrfsTer7 transfected cells (Fig 3A).

In *RelA*$^{-/-}$ MEF cells the ratio of cleaved caspase 3/caspase 3 induced by TNF exposure was 1.98-fold higher (p=0.036) in RELAp.His487ThrfsTer7 compared to RELA^{WT} (Fig 3B). In PBMCs, we compared expression of two anti-apoptotic markers that had previously been associated with RELA haploinsufficiency, TRAF1 and BCL2A1(5). TRAF1 gene expression was upregulated 7.62-fold in symptomatic RELAp.His487ThrfsTer7 heterozygotes compared to WT familial controls (p=0.005, Fig 3C). Exposure to TNF down-regulated TRAF1 2.7-fold (+/-0.591, p=0.045) compared

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to unstimulated controls. BLC2A1 was upregulated 22.2-fold in RELAp.His487ThrfsTer7 patient cells compared to their WT familial controls (p=0.049, Fig 3D).

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DISCUSSION

LOF variants in two NF-κB pathway genes, *TNFAIP3* and *RELA*, have been implicated in monogenic forms of Behçet's-like autoinflammatory syndromes(5, 9). Here we describe the first reported case associated with RELA truncation, rather than haploinsufficiency. Both protein variants were expressed at approximately equal levels in PBMCs. Mutant proteins have the potential to modulate biological systems in ways distinct from RELA loss. The mutation causes deficiencies of NF-κB signalling leading to enhanced TNF-induced apoptosis.

Heterozygous individuals had increased levels of RELA and TNFAIP3, perhaps related to disease activity, however there was loss of TNF-induced upregulation of these genes. NF-κB modulation of gene expression is determined by a range of factors including graded dose discrimination and NF-κB oscillations (10). We demonstrated in a fibroblast model that the mutant allele is a much weaker transcriptional activator compared to wild-type RELA unrelated to the kinetics of translocation of RELA variants to the nucleus. This is likely to be due to loss of DNA binding activities or co-factor binding that are known to reside in this particular C-terminal region(11). Although the mutation does not directly involve the DNA binding domain of RELA, distal residues that do not directly contact DNA can affect DNA:RELA proximity and electrostatic surface potential, which can alter DNA:RELA recognition and binding affinity (12). Thus, the RELAp.His487ThrfsTer7 mutant appears to behave as a hypomorph, maintaining certain key elements of RELA function, e.g. the ability to translocate to the nucleus in response to TNF but losing key features associated with its C-terminal transactivation domain. A similar observation was made in a single individual with a de novo missense RELA mutation (c.1534_1535delinsAG, p.Asp512Ser) who had impaired NF-κB-responses in LPS stimulated fibroblasts(13). .

Previously, a two generation USA family with an autosomal-dominant BD-like condition with a reported with a *RELA* variant in the canonical spice donor site was reported (5). In this family, the mutant transcript was not detectable with consequent RELA haploinsufficiency. The associated illness included mucocutaneous ulceration and early onset colitis responsive to anti-TNF therapy. The pathogenic mechanism is explained by increased sensitivity of mutant fibroblasts to TNF-induced apoptosis, which we replicated in our study using transfected MEF cells. A recently

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described autosomal dominant autoinflammatory condition due to non-cleavable RIPK1 is also explained by hypersensitivity to apoptosis and necroptosis in myeloid cells while fibroblasts of these patients appear to be resistant to cell death(14). The later onset of ulceration and lack of colitis in our RELAp.His487ThrfsTer7 family may be related to a less severe reduction in protein function compared with the 50% reduction in the US family. RELA is a key regulator of homeostasis at the surface of the mucosa, promoting epithelial barrier integrity. Once this barrier is compromised, a genetic impairment in RELA-dependent signalling may lead to abnormal sensitivity to TNF.

It is noteworthy that a family member with the mutation had an autoimmune condition, NMO, without features of the BD-like syndrome. Both BD and NMO are relapsing and remitting inflammatory disorders and may share similar pathophysiology(6). Furthermore, optic neuritis is a recognised if uncommon clinical feature of BD. The individual with NMO had no clinical or imaging evidence of other nervous system involvement such as transverse myelitis or cerebral lesions. The co-occurrence of autoimmune and autoinflammatory conditions has not been widely reported, however a review of TNFAIP3-related autosomal dominant BD found a high prevalence of autoantibodies associated with common autoimmune conditions including systemic lupus erythematosus, type 1 diabetes and Hashimoto's thyroiditis. Notably, common variants in *TNFAIP3* have been associated with risk of most of these conditions(15).

Autosomal dominant forms of BD-like autoinflammation related to *TNFAIP3* LOF mutations are increasingly recognised. The pathogenic mechanism is believed to involve increased inflammasome activation and production of pro-inflammatory cytokines by leukocytes. Our data demonstrates a mechanism of increased sensitivity to TNF-induced apoptosis caused by LOF mutations in RELA. The emerging mechanistic and clinical studies suggest identification of the underlying genetic causes of Mendelian forms of BD could inform a personalised medicine approach using either IL-1 or TNF inhibitors.

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LEGENDS TO FIGURES

Figure 1. Pedigree of family with mucocutaneous ulcerative condition.

(A) The genotype of RELA c.1459delC. White are homozygous wildtype, grey and heterozygotes. Patterned shading represents the individual heterozygous for RELA c.1459delC with NMO. Circles represent females, squares represent males. (B) Sanger sequencing of *RELA* showing a wild-type familial control and a heterozygous affected individual with cytosine base deletion (arrow) at position c.1459. (C) Graphical representation of full length wild-type (WT) RELA with conserved Rel homology domain (RHD) and two C-terminal transactivation domains (TA2 &TA1), and RELAp.His487ThrfsTer7 with partial TA2 but lacking TA1.

Figure 2. RELAp.His487ThrfsTer7 is an expressed truncated protein and is associated with altered NF-κB signalling.

(A) Individuals heterozygous for *RELA*c.1459delC have two detectable RELA bands when probed with a RELA N-terminus, but not C-terminus antibody. The RELA bands correlate to full length RELA and the predicted RELAp.His487ThrfsTer7. (B) RELA, NFKBIA and TNFAIP3 mRNA is upregulated in RELAp.His487ThrfsTer7 patient cells compared to WT familial control PBMC cells. Graphs show mean Ln(relative quantitation (RQ)) for each individual. TNF stimulation was 4 hours. (C) Luciferase activity in MEF *RelA*$^{-/-}$ transfection model demonstrates the impaired ability of RELAp.His487ThrfsTer7 to transactivate NF-kB response elements. Results show average fold increase of NF-κB activity in TNF stimulated cells compared to unstimulated pcDNA, plotted as luc/ren. (D) Representative confocal images of MEF *RelA*$^{-/-}$ transfection models treated with TNF for the indicated times. (RELA is green. Nucleus is blue. Scale bar 10µm. Bar graphs represent mean ± SEM unless stated otherwise, n=3 independent experiments (* $p<0.05$; **$p<0.01$, ***$p<0.001$)).

Figure 3: RELAp.His487ThrfsTer7 increases apoptosis

(A) Western blot and densitometry of HEK293T transfection model cells shows no difference in expression levels of RELA^{WT} and RELAp.His487ThrfsTer7 Cells expressing RELAp.His487ThrfsTer7

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show increased BAX and decreased BCLX, both basally and in response to TNF, compared to $RELA^{WT}$ (NTC; Non-transfected cells). BCL2 is downregulated by TNF but is not significantly different between RELAp.His487ThrfsTer7 and $RELA^{WT}$ expressing cells. (B) TNF induces an increased ratio of cleaved caspase 3 (compared to pro-caspase 3) in cells expressing RELAp.His487ThrfsTer7 compared to $RELA^{WT}$. (C, D) *TRAF1* and *BLC2A1* mRNA is upregulated in RELAp.His487ThrfsTer7 patient PBMC cells compared to WT familial control PBMCs. TNF induces a downregulation of *TRAF1* in RELAp.His487ThrfsTer7

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Ethics: Approval was obtained from the Limerick University Hospital Medical Research Ethics Committee in accordance with the Declaration of Helsinki principles.

PPI involvement: The UCD Centre for Arthritis Research is governed by a steering committee which includes 3 patient representatives with 1/3 of the voting power. The steering committee oversees all research activities of the Centre. Direct patient involvement activities for this study focus on the dissemination of the study outputs to public stakeholders. This study has a patient research partner who is a member of the project advisory group, mentors the researchers in lay communication, and advises on our patient consent forms.

Competing interest: none.

Contributions: Conceived study: ED, AGW, FA, EC, OK, CT, SS, AF. Experimental design: AGW, ED, EC, DL. Clinical data and sample collection: FA, AQ, WLN, AF. Performed experiments and data analysis: ED, NM, DL. All authors contributed to the preparation of this manuscript and read and approved the final version.

KEY MESSAGES

- Truncation in RELA is associated with both Behcet's-like disease and Neuromyelitis Optica in an Irish family.
- Pathogenecity is associated with loss of RELA transactivation domains resulting in increased apoptosis and altered NF-κB signalling

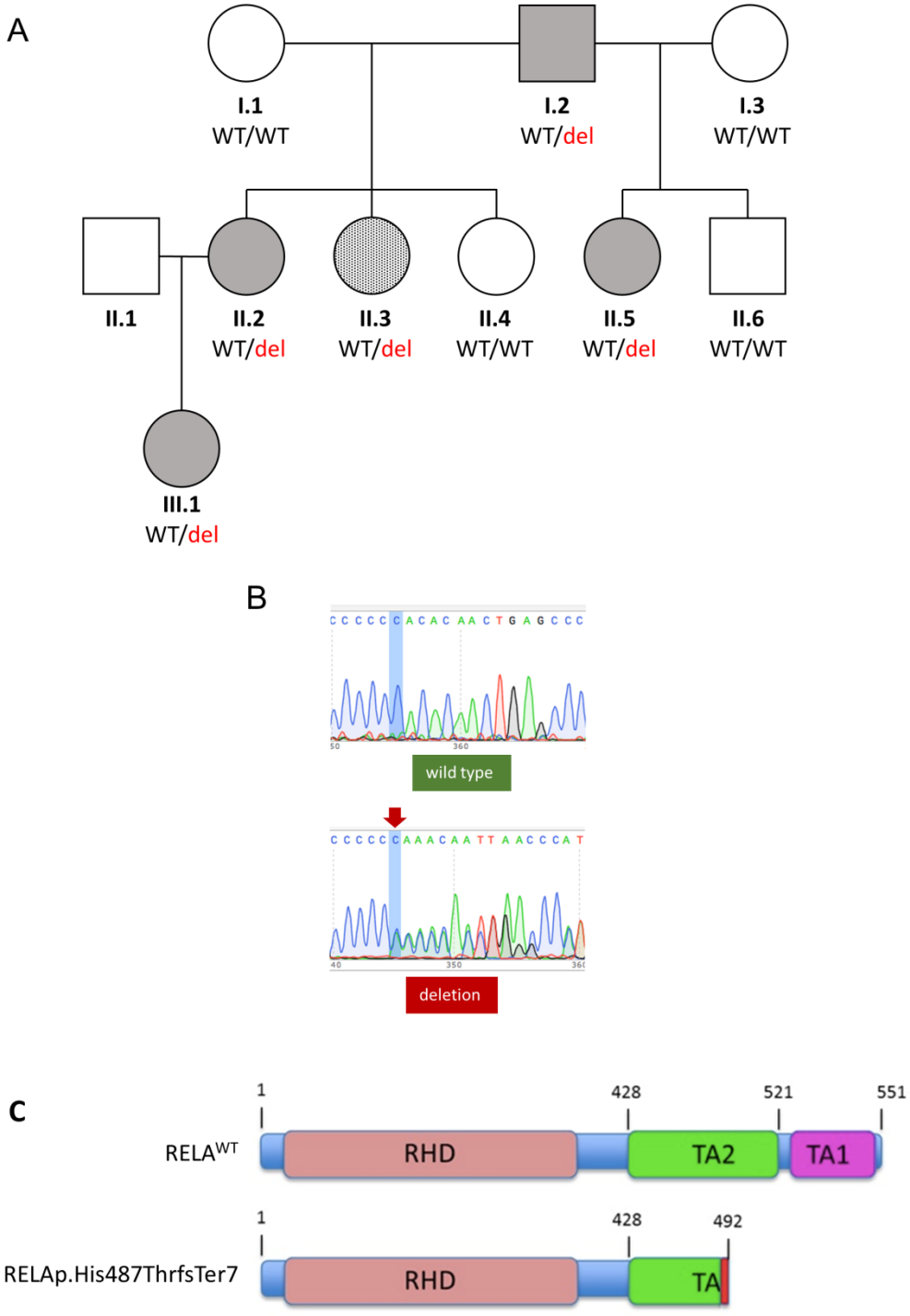
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- This data supports a growing body of evidence that the NF-κB pathway is central in monogenic BD-like autoinflammation.

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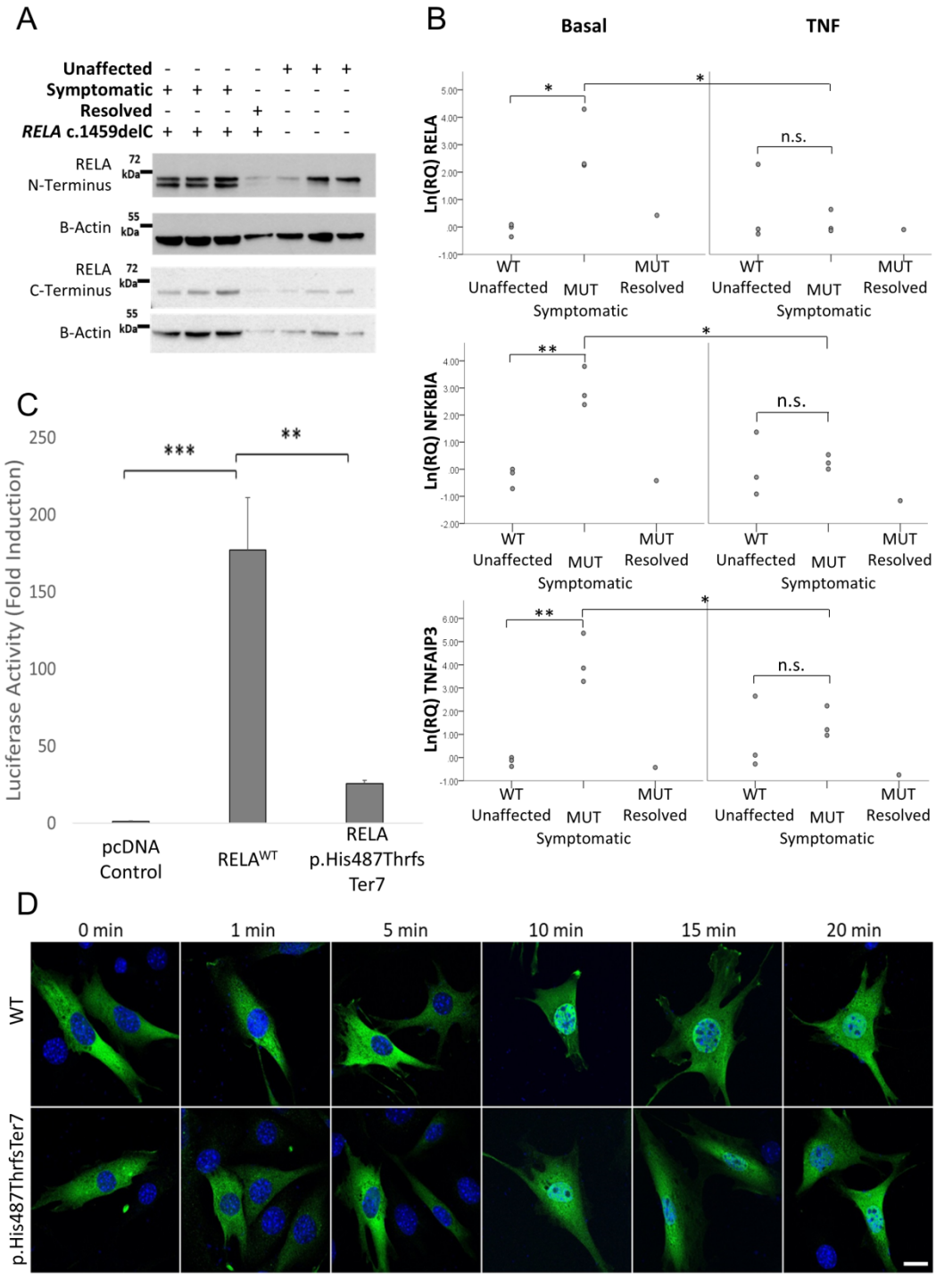
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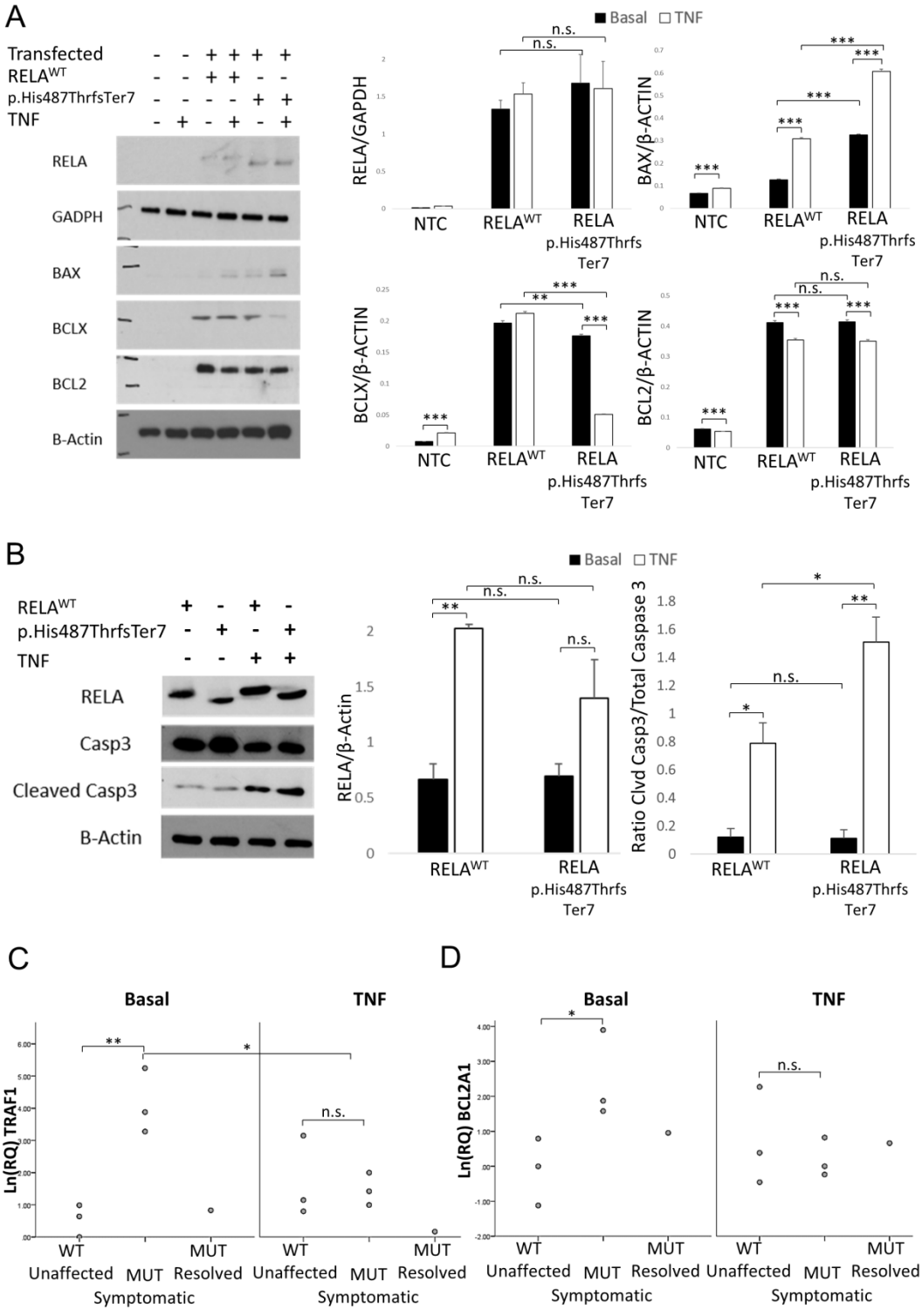
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